LAMMPS official dem03-legacy-hertzhistory-wall

Hertz contact energy at the source check time | 29 October 2026

PASS	Four main runs evaluated using the unchanged official selected-time analytic criteria; source tag unvetted.

Official quantities, errors and acceptance limits

Quantity	Reference / limit	CPU dt	GPU dt	CPU dt/2	GPU dt/2
Hertz peak energy (J)	0.003484223 relative limit 0.01	0.003484224 err 3.318e-07 PASS	0.003484224 err 3.318e-07 PASS	0.003484223 err 3.973e-08 PASS	0.003484223 err 3.973e-08 PASS

Check time=2.138e-05 s; original step 2138. Half dt uses twice the check step at the same time. Each cell shows the native value and its own absolute/relative error; a reference limit is not a same-code epsilon.

Original bodies and source-to-native mapping

Body	Radius / density	Initial xyz (m)	Initial v (m/s)	Initial spin (rad/s)
sphere	R=0.0025 rho=7000	(0, 0, 0.00255)	(0, 0, -3.9)	(0, 0, 0)

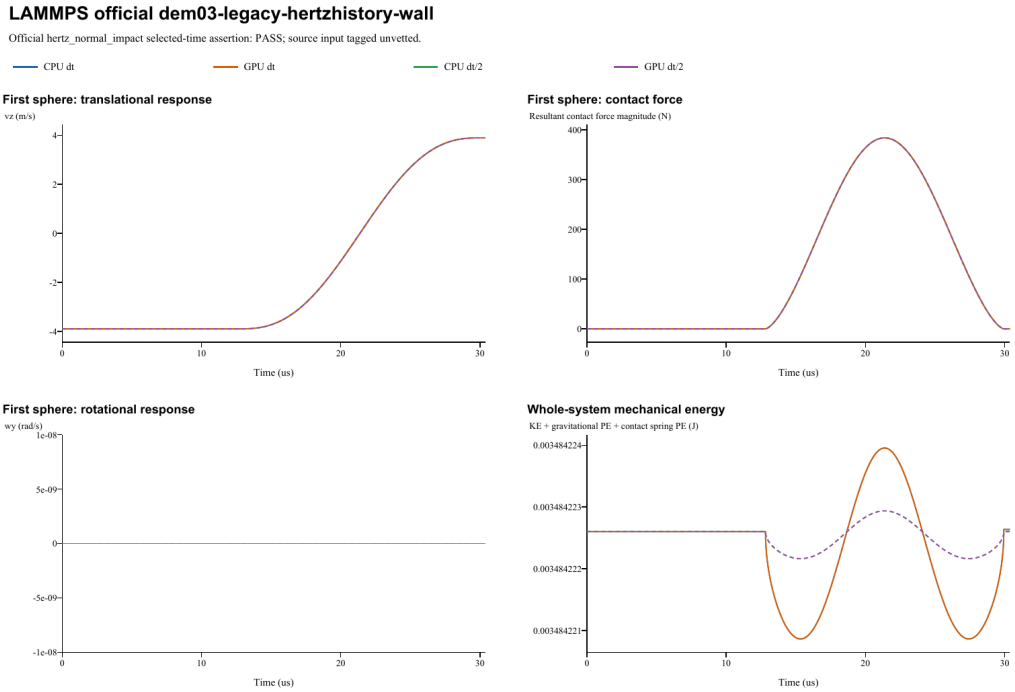
Hertz Fn_elastic=(4/3)*E_eff*sqrt(R_eff)*delta^1.5; native E_eff=5.3333325e+10 Pa. Stored linear kn is inactive. Automatic damping uses the harmonic material restitution and the active normal/tangential stiffness; stored manual dashpots are inactive. Tangential inputs: kt=0 N/m (explicit in linear mode), mu=0. Rolling/twist coefficients: kr=0, mu_r=0, ktwist=0, mu_twist=0.

Gravity=(0, 0, 0) m/s^2. Base/half dt=1e-08/5e-09 s; 3038/6076 steps. Legacy Hertz zero normal damping maps to native e1 and E*=3kn/4. The source NULL tangential Hertz stiffness is dormant because relative tangential velocity is identically zero. This verifies only the normal degeneration.

Pinned LAMMPS source

Official input | Analytic checker | Parameter mapping.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; input blob 9b8a5577c5c27f8beacb2402173c0c68df6fb6f8. Official tag: unvetted.
LAMMPS was not rerun locally. Physical inputs were not fitted to native results.

LAMMPS native histories and refinement



CPU/GPU and half-step comparison

Quantity	CPU dt error	CPU dt/2 error	Error type / limit
Hertz peak energy (J)	3.3176e-07	3.9731e-08	relative: 0.01

CPU/GPU whole-history maximum position / velocity / spin / force differences: 0 m / 8.88e-16 m/s / 0 rad/s / 0 N.

Maximum individual impulse / position-drift / velocity-kick residuals: 1.6e-17 kg m/s / 2.17e-19 m / 4.38e-16 m/s. Gravity is added separately to saved native contact forces.

Complete results and verified scope

All 18,232 native frames are retained; main frame counts=[3039, 3039, 6077, 6077]. Every numerical-step particle/contact CSV, four loadable main projects and raw initial samples remain preserved. Output cadence is recorded separately from integration dt. Energy/history diagnostics do not replace the official acceptance quantities.

Run manifest | Analysis | Official metrics | Independent audit | Full native ZIP | Frame hashes

Source checkpoints: Source checkpoint states are supplemental regression diagnostics. Their differences are reported directly; the YAML same-code epsilon is not imposed as an independent cross-software acceptance limit.

A stable helper executable is statically linked to the current native CPU/CUDA libraries. Historical GUI/library byte identity is not asserted. Scope is the active source physics and the selected official criteria; inactive options remain outside this test.